Project Planning & Management

SOC Automation Lab for Threat Detection and Incident Response

Project Overview

This project aims to build a Security Operations Center (SOC) lab capable of detecting cyber attacks, analyzing logs, and automating incident response using open-source security tools.

Objectives

• Detect malicious activity on endpoints

• Centralize logs using a SIEM platform

• Automate threat intelligence enrichment

• Automatically generate incidents

• Provide SOC monitoring visibility

Project Timeline

Phase	Duration	Description
Planning	Week 1	Define project scope
System Design	Week 2	Architecture and diagrams
Implementation	Week 3-4	Deploy security tools
Integration	Week 5	Integrate systems
Testing	Week 6	Validate detection

Project Milestones

Milestone	Description
Architecture Completed	SOC architecture finalized
Infrastructure Ready	All virtual machines deployed
Tools Installed	Security tools installed and configured
Integration Completed	All systems integrated
Testing Completed	Attack simulations successful

Project Deliverables

Deliverable	Description
Project Documentation	Full technical report
SOC Lab Environment	Configured virtual environment
Detection Demonstration	Attack detection demo
GitHub Repository	Source code and configs

Resource Allocation

Resource	Purpose
Virtual Machines	Run lab infrastructure
Windows Endpoint	Generate logs
SIEM Server	Log analysis platform
Automation Server	Run automation workflows
Incident Response Platform	Manage incidents

Risk Assessment

Risk	Mitigation
VM resource limitation	Allocate sufficient RAM/CPU
Integration failure	Test components individually
Log overload	Implement log filtering

KPIs

KPI	Description
Detection Time	Time required to detect an attack
Alert Accuracy	Percentage of valid alerts
Incident Creation Time	Time to generate incident
System Availability	SOC system uptime